

Task 2 Report – Jack Millington s5405915

1.0 Introduction

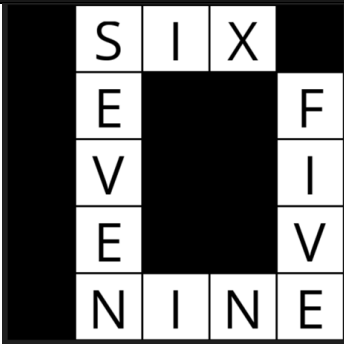
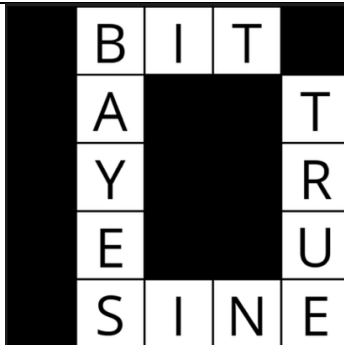
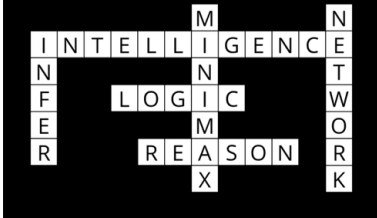
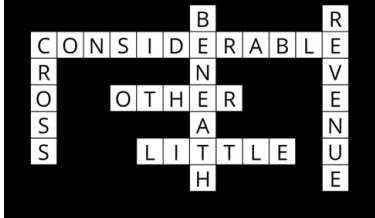
This report summarises the results from a Crossword CSP puzzle. This was tested using 3 different crossword structures, and 3 different word banks, totalling to 9 separate experiments. This was written in python, running on the provided generate.py, and uses AC-3 and backtracking.

The number of variables in each structure and words in each word bank is shown below.

Structure:	Variables:	Word Bank	Words:
Structure0	4	Words0	10
Structure1	6	Words1	51
Structure2	6	Words2	3000

2.0 Results

The following table shows the output of each structure tested with 3 different word banks. It also shows the number of backtrack calls and time taken for diagnostics.

Structure0, Words0 Backtrack Time taken: 0.000082s Backtrack calls 5 	Structure0, Words1 Backtrack Time taken: 0.000084s Backtrack calls 5 	Structure0, Words2 Backtrack Time taken: 0.0184s Backtrack calls 5 
Structure1, Words0 No solution.	Structure1, Words1 Backtrack Time taken: 0.0002s Backtrack calls 7 	Structure1, Words2 Backtrack Time taken: 0.0176s Backtrack calls 7 

Structure2, Words0	Structure2, Words1	Structure2, Words2																																																																								
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No solution.	<table><tr><td></td><td></td><td></td><td></td><td></td><td>C</td></tr><tr><td>L</td><td>O</td><td>S</td><td>S</td><td></td><td>R</td></tr><tr><td>I</td><td></td><td></td><td>T</td><td>R</td><td>U</td></tr><tr><td>N</td><td></td><td></td><td>A</td><td></td><td>A</td></tr><tr><td>E</td><td></td><td></td><td>R</td><td></td><td>T</td></tr><tr><td></td><td>B</td><td>I</td><td>T</td><td></td><td>E</td></tr></table>						C	L	O	S	S		R	I			T	R	U	N			A		A	E			R		T		B	I	T		E	<table><tr><td></td><td></td><td></td><td></td><td></td><td>T</td></tr><tr><td>S</td><td>A</td><td>M</td><td>E</td><td></td><td>W</td></tr><tr><td>H</td><td></td><td></td><td>L</td><td>A</td><td>K</td></tr><tr><td>U</td><td></td><td></td><td>I</td><td></td><td>N</td></tr><tr><td>T</td><td></td><td></td><td>T</td><td></td><td>T</td></tr><tr><td></td><td>I</td><td>C</td><td>E</td><td></td><td>Y</td></tr></table>						T	S	A	M	E		W	H			L	A	K	U			I		N	T			T		T		I	C	E		Y
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3.0 Analysis

It can be extrapolated that the number of backtrack calls are directly related to the number of variables. Where:

$$\text{Number of Backtrack Calls} = \text{Number of Variables} + 1$$

It can also be seen that the Backtrack function time taken is heavily related to the number of words in the bank. With the number of variables also having some aspect in this, due to the increased number of backtracking calls. Word bank growth has a much stronger effect than variable growth in these experiments, however, the variable sizes did not vary much compared to the word bank sizes. Some combinations give no solution, AC-3 detects inconsistency early, so this is useful as it saves wasted search time. The MRV and degree heuristic improve search efficiency as it chooses the more promising option, cutting down on time.

Some limitations include that the current solver doesn't enforce uniqueness until consistent, which could waste time exploring duplicate values. Also, speed could be increased further by maintaining arc consistency during backtracking, interleaving search with inference. Also, word banks could be pre-sorted by length, which could also cut on time.

4.0 Conclusion

Overall, the crossword CSP solver performed well on small structures and word banks, consistently finding solutions with a small number of backtrack calls. The experiments determined that runtime is mainly driven by the size of a word bank, with the number of variables determining the number of recursive calls. The system correctly determined no solution early, eliminating wasted time. The MRV, degree, and LCV heuristics cut down on run time, increasing efficiency of the program. These findings highlight the strengths of constraint propagation, node and arc consistency, combined with backtracking, while also showing scalability limits and possible improvements for larger problem instances.